RSBuilding: Toward General Remote Sensing Image Building Extraction and Change Detection With Foundation Model

Mingze Wang, Lili Su, Cilin Yan, Sheng Xu, Pengcheng Yuan, Xiaolong Jiang, and Baochang Zhang, *Senior Member, IEEE*

***Abstract*— Buildings not only constitute a significant proportion of man-made structures but also serve as a crucial component of geographic information databases, closely linked to human activities. The intelligent interpretation of buildings plays a significant role in urban planning and management, macroeconomic analysis, population dynamics, etc. Remote sensing image building interpretation primarily encompasses building extraction and change detection (CD). However, current methodologies often treat these two tasks as separate entities, thereby failing to leverage shared knowledge. Moreover, the complexity and diversity of remote sensing image scenes pose additional challenges, as most algorithms are designed to model individual small datasets, thus lacking cross-scene generalization. In this article, we propose a comprehensive remote sensing image building understanding model, termed RSBuilding, developed from the perspective of the foundation model. RSBuilding is designed to enhance cross-scene generalization and task universality. Specifically, we extract image features based on the prior knowledge of the foundation model and devise a multilevel feature sampler to augment scale information. To unify task representation and integrate image spatiotemporal clues, we introduce a cross-attention decoder with task prompts. Addressing the current shortage of datasets that incorporate annotations for both tasks, we have developed a federated training strategy to facilitate smooth model convergence even when supervision for some tasks is missing, thereby bolstering the complementarity of different tasks. Our model was trained on a dataset comprising up to 245 000 images and validated on multiple building extraction and CD datasets. The experimental results substantiate that RSBuilding can concurrently handle two structurally distinct tasks and exhibits robust zero-shot generalization capabilities. The code will be made available for open-source access at https://github.com/Meize0729/RSBuilding.**

***Index Terms*— Building extraction, change detection (CD), federated training, foundation model, remote sensing images.**

Manuscript received 15 March 2024; revised 18 June 2024; accepted 23 July 2024. Date of publication 6 August 2024; date of current version 12 September 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFC3300029; in part by Zhejiang Provincial Natural Science Foundation of China under Grant LD24F020007; in part by Beijing Natural Science Foundation under Grant L223024; in part by the National Natural Science Foundation of China under Grant 62076016 and Grant 12201024; in part by the "One Thousand Plan" Projects in Jiangxi Province under Grant Jxsg2023102268; in part by Beijing Municipal Science and Technology Commission, Administrative Commission of Zhongguancun Science Park, under Grant Z231100005923035; and in part by Taiyuan City "Double Hundred Research Action" under Grant 2024TYJB0127. *(Corresponding author: Baochang Zhang.)*

Mingze Wang, Cilin Yan, and Sheng Xu are with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China.

Lili Su is with the State Key Laboratory of Networking and Switching Technology, Beijing University of Posts and Telecommunications, Beijing 100876, China.

Pengcheng Yuan and Xiaolong Jiang are with Xiaohongshu Inc., Beijing 100020, China.

Baochang Zhang is with Hangzhou Research Institute, Institute of Artificial Intelligence, Beihang University, Beijing 100191, China, also with Zhongguancun Laboratory, Beijing 100190, China, and also with Nanchang Institute of Technology, Nanchang 330044, China (e-mail: bczhang@buaa.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3439395

I. Introduction

BUILDINGS serve as the primary components of man-made surface features, embodying a crucial aspect of specialized geographic information systems that are inherently associated with human activities [1]. The advent of remote sensing technology has significantly streamlined the procedure of mapping building distribution across broad spatial and temporal scales. Techniques for building extraction and change detection (CD) play a fundamental role in interpreting building-related data from remote sensing imagery. These methods offer an optimal and efficient pathway for urban planning and management, as well as macroeconomic analysis [1], [2], [3], [4], [5], [6].

Owing to significant advancements in computational capabilities and model development [7], [8], deep learning technology has progressively become the predominant approach for extracting building-related information from remote sensing imagery [1], [9]. Specifically, deep learning-based technologies for building footprint extraction and CD offer the capacity to interpret single-temporal and multitemporal remote sensing images, respectively. The former concentrates on the delineation, distribution, and area of buildings, while the latter underscores the construction and demolition of buildings from a diverse range of spatiotemporal remote sensing data [10]. Importantly, both tasks converge on the same research subject: the building.

Currently, most methods are specifically designed for individual tasks. Techniques for building extraction strive to overcome challenges such as low segmentation accuracy, blurred edges, small targets, varying image scales, etc. These techniques employ structures such as dual-stream networks, multiregion attention, boundary optimization, and vectorization to augment segmentation results [1], [11], [12], [13]. The capacity of the transformer [14] to model long-range dependencies has further catalyzed the evolution of building extraction algorithms, thereby offering more generalized scene adaptability. STT [1] establishes correlations between

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

buildings with minimal cost by leveraging a sparse attention mechanism in both space and channels, thereby achieving superior segmentation performance. BuildFormer [12] tackles the issue of spatial details loss by ingeniously designing a spatial-detailed context path for encoding rich spatial details, alongside a global context path tailored to capturing global dependencies.

CD primarily concentrates on the changes of the buildings. This process not only grapples with the inherent challenges of extracting buildings from single-temporal remote sensing images but also emphasizes the fusion of features across multitemporal images. Early research predominantly centered around the analysis of differences in input or output, eventually evolving toward feature extraction for the construction of classifiers. However, these methodologies proved insufficient in addressing the complexities of remote sensing image scenes [15], [16]. Contemporary techniques aim at multitemporal feature fusion, which can be divided into pre-fusion and post-fusion [3], [5]. Pre-fusion incorporates temporal information during the feature extraction phase in the backbone, while post-fusion typically gathers temporal information after the spatial features of each image have been extracted. Both strategies harbor their unique advantages and have demonstrated remarkable results. With the emergence and growing popularity of the transformer model, BiT [3] has ventured into the utilization of the attention mechanism for the fusion of bi-temporal features. ChangeFormer [17] has established a spatiotemporal correlation by constructing a pure attention feature extraction network. Both have achieved state-of-the-art performance in their respective periods.

Substantial progress in foundation models has paved the way for a more generalized interpretation of remote sensing images. Within the domain of computer vision, CLIP [18] has been introduced to align the representational space of images and text by utilizing noise contrastive estimation (NCE) loss. Conversely, SAM [19] employs a training-annotation loop data engine to conduct supervised training on extensive annotations (11 million images, 1 billion masks), culminating in what is currently the most comprehensive segmentation model. RSPrompter [6], a model grounded in prompt learning and built upon the SAM [19] foundation model, can automatically extract instance-level building masks with minimal learning cost. The BAN [2] integrates the knowledge of the foundation model into the CD task through a dual-temporal adapter network. TTP [5] introduces an innovative spatiotemporal activation gate and a multiscale enhancer, thereby enabling the visual foundation model to leverage multitemporal information and excel in CD tasks.

Despite significant advancements in the interpretation of buildings from remote sensing imagery, contemporary methodologies predominantly segregate the tasks of building extraction and CD [3], [12], [20]. These approaches overlook the potential benefits of leveraging shared knowledge to enhance performance [21]. Furthermore, the intricate scenes and diverse resolutions inherent in remote sensing images present substantial challenges [22], [23]. Existing algorithms concentrate on modeling individual datasets in isolation and configuring hyperparameters independently, thereby constraining their generalizability across multiple scenes and scales [2], [17]. While there have been developments in algorithms based on foundation models, these methods primarily depend on the general knowledge derived from natural image interpretation tasks. They are fine-tuned and transferred on individual datasets and are far from practically applying to the task of interpreting buildings from remote sensing images.

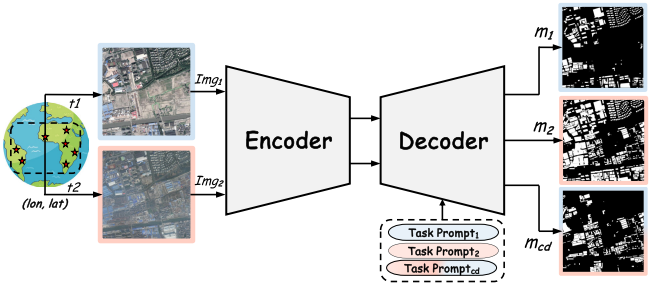


Fig. 1. Proposed foundation model for remote sensing image building understanding. It possesses the capability to simultaneously output building segmentation masks and change masks across different spatial and temporal points, demonstrating strong generalization ability.

Additionally, single-variable CD can only provide binary information regarding the presence or absence of change. Incorporating a building extraction branch allows for a combined analysis that elucidates the nature of the changes, such as demolition or new construction. Many applications also require access to both CD results and building extraction data simultaneously. For instance, in urban planning, agencies monitor building changes over time to understand population and economic shifts. By comparing building data from various times, they get clearer insights on building layouts and density, enhancing development evaluation and future planning. For environmental protection, authorities track building changes in protected areas to assess human impact on nature. Access to detailed building information at different times allows for precise environment analyses.

In this article, our aim is to integrate the tasks of building extraction and CD within a single framework, while also investigating their mutual benefits. To achieve this, we present a general model for remote sensing image building understanding, which is rooted in the foundation model. This approach frees researchers from the constraints of detailed subbranches or submodules, thus improving task universality and generalization across various scenarios during interpretation, as shown in Fig. 1.

To achieve this, we introduce RSBuilding, a foundation model designed for both building extraction and CD. Specifically, we introduce a robust backbone with a large capacity for knowledge for remote sensing imagery, based on the transformer encoder, which can extract more general and refined features from dual images. Simultaneously, we have designed a streamlined multilevel feature enhancer to augment scale information. This is aimed at overcoming challenges presented by variability in size, color, and overall information of buildings in remote sensing images, along with issues related to mutual occlusion and difficulty in edge extraction.

To unify the representations of different tasks and amalgamate associated dual-image features, we propose a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

cross-attention decoder that operates based on task prompts. Furthermore, to address the existing scarcity of datasets that provide comprehensive annotations for both tasks, we develop a federated training strategy. This strategy facilitates the smooth convergence of the model even in the absence of supervised information for certain tasks, thereby enhancing the complementarity between the two individual interpretation tasks.

The primary contributions of this article can be summarized as follows.

1) We propose a general foundation model for building interpretation from remote sensing images, integrating building extraction and CD tasks within a unified framework, thereby highlighting and leveraging the synergistic relationship between these two tasks.
2) We introduce a streamlined multilevel feature sampler to construct a feature pyramid, enriching the multiscale features of buildings. We also propose a cross-attention decoder based on task prompts to aggregate task-specific features and generate the corresponding mask. Additionally, we develop a federated training strategy to address the issue of missing part annotations in multitask learning.
3) We have amassed and structured a building dataset consisting of approximately 245 000 images for multitask training. We validated our pre-trained foundation model on several key datasets, including WHU and INRIA for building extraction, LEVIR-CD and S2Looking for CD, and BANDON for both tasks. Experimental results indicate that our proposed method is capable of handling two structurally diverse tasks simultaneously and demonstrates robust generalization capabilities across a variety of application scenes.

The remainder of this article is organized as follows: Section II presents a comprehensive review of the relevant literature. In Section III, we delve into the specifics of the structure and training strategy of our proposed RSBuilding. Section IV introduces the datasets and the evaluation protocols and provides an in-depth analysis of both quantitative and qualitative results. This section further includes ablation studies, supplemented by a thorough discussion and an acknowledgment of limitations. Finally, Section V concludes the article by encapsulating the key findings and insights.

II. Related Works

A. Building Extraction

Remote sensing image building extraction involves pixel-level classification to generate binary masks indicating the presence or absence of buildings, which can be conceptualized as a semantic segmentation problem [11]. As one of the fundamental tasks in remote sensing image interpretation, it has broad applications in urban planning, map services, and disaster management. Historically, building extraction methods primarily employed traditional machine learning techniques such as contour extraction, edge gradients, and gray-level co-occurrence matrices [24], [25], [26], [27], [28], [29], [30], etc. However, these methods often struggled to address challenges related to varying scale ranges and shapes, and were also characterized by complex procedures with numerous hyperparameters [13].

Deep learning [31], [32] presents a more effective approach to these challenges, yielding more accurate segmentation results. Early deep learning-based semantic segmentation methods predominantly utilized the fully convolutional network (FCN) [33] architecture, which employs full convolution operations to extract high-level semantic features and deconvolution operations to restore the feature to the original image size. For instance, MA-FCN [34] proposed a multiscale fusion-based FCN for building extraction, demonstrating robustness across scales. Subsequently, encoder–decoder structures emerged as the mainstream solution for segmentation tasks. UNet [35] architecture, which first showed excellent performance in medical image segmentation through its fully convolutional encoder–decoder architecture and skip connections between feature maps of different scales, was adapted for various fields, including building extraction. Meanwhile, the KGLPD [36] framework incorporates adaptive gradient perceptive, land pattern cognitive models, and a fully sparse topic model to improve land-use mapping accuracy. However, the limited receptive field of convolution operators remained a significant constraint in obtaining global semantic information. The advent of the transformer structure led to the development of segmentation networks with attention structures, such as SegFormer [37], which showed remarkable performance. For example, STT [1] proposed a convolution-transformer hybrid structure with a sparse attention mechanism, striking a balance between accuracy and efficiency. BuildFormer [12] used a spatial context path to encode rich spatial details and a global context path to capture global dependencies. Recently, tackling building extraction's generality and transferability, SKTrans [38] integrates a tone shift mechanism and knowledge module, addressing regional tone variations and stylistic diversity in large-scale, unlabeled datasets.

Despite these transformer-based methods achieving impressive results on specific datasets, their generalizability across datasets and scenes remains limited, and they struggle to maintain stable performance in complex and diverse remote sensing scenarios. In this article, we aim to enhance the model's general and generalizable capabilities in understanding buildings through the federated training of building segmentation and building CD tasks.

B. Change Detection

CD, a critical component in the analysis of multitemporal remote sensing images, is designed to identify variations within specific regions or categories [3], [9]. These variations, which may present as either an expansion or contraction of geographical features, yield dynamic insights into human activities and regional economic shifts. Currently, the primary focus of CD tasks is the monitoring of building alterations due to their significant correlation with human life and economic activities [2].

Historically, CD methodologies were predominantly centered around feature engineering. These methods entailed the construction of multitemporal features, followed by pixel or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

region-based classification. However, these approaches were hindered by several limitations, including limited parameter capacity, inadequate adaptability, and restricted generalization capabilities [2], [5], [39].

Recent advancements in deep learning have precipitated a shift toward network-based CD methods. Leveraging the capabilities of neural networks, these methods are capable of extracting robust image features and performing spatiotemporal fusion across various feature extraction stages effectively [16], [40]. Early deep learning-based algorithms perceived CD as a dense prediction task, with the majority exploring different stages of bi-temporal feature fusion to generate the final change segmentation map. For instance, Daudt et al. [41] proposed three typical CD models: FC-EF, FC-Siam-Conc, and FC-Siam-Diff, which generate high-precision change masks by fusing features from different periods. STANet [9] introduced an innovative Siamese-based spatial–temporal attention neural network, incorporating a self-attention mechanism to effectively model spatial–temporal relationships. This addresses the challenge of capturing spatial–temporal dependencies at various scales.

Recently, the advent of transformer architectures has introduced new possibilities for the integration of spatiotemporal features in CD tasks. BiT [3], combining convolutional neural network (CNN) and transformers, establishes an efficient paradigm with a minimal parameter footprint for the fusion of multitemporal features. ChangeFormer [17], a model that exclusively relies on transformer-based Siamese networks and fully connected segmentation heads, demonstrates exceptional performance. Changer [42] verifies the efficacy of feature interactions in CD by capitalizing on the contextual information derived from paired temporal features.

In this article, we present a transformer-based network specifically designed for enhancing building comprehension. By employing federated learning and investigating general knowledge, we aim to augment the precision of building CD and expand the application scenarios of the proposed model.

C. Foundation Model

The advent and evolution of foundation models have catalyzed the transformation of artificial intelligence, facilitating the implementation and advancement of more generalized and human-like methodologies across a multitude of application contexts, thereby yielding substantial commercial values. Foundation models, also known as, base models, predominantly comprise deep learning models characterized by an extensive number of parameters. Owing to their self-supervised or semi-supervised training on voluminous data, these models are capable of quickly adapting to a diverse range of downstream tasks, demonstrating remarkable zero-shot inference capabilities [18], [19], [43].

The early iterations of foundation models primarily consisted of pre-trained language models, such as Google's BERT [44] and OpenAI's GPT [18], [45], [46], [47] series, facilitated by the availability of a vast corpus. These large language models can be rapidly deployed to specific scenarios through fine-tuning, in-context learning, and other techniques [48], [49]. The introduction of the language dialog model, ChatGPT [50], and the multimodal understanding model, GPT-4 [47], has effectively initiated research into more versatile foundation models.

Subsequently, a variety of visual foundation models and multimodal foundation models have emerged, including stable diffusion [51], Flamingo [52], SAM [19], etc. MiniGPT4 [53], by integrating an image encoder with a large language model, can adapt to a wide range of multimodal visual understanding tasks, demonstrating exceptional performance. SAM [19], leveraging the proposed data annotation engine, trains on up to 1 billion masks, thereby exhibiting comprehensive segmentation capabilities.

In the realm of remote sensing, the majority of contemporary methods adhere to the paradigm of fine-tuning models on downstream tasks by extracting high-level semantic knowledge from remote sensing images or geospatial spaces. For instance, RingMo [54] acquires a general representation of remote sensing images through the unsupervised image reconstruction training strategy of masked image modeling (MIM) on 2 million remote sensing images. A handful of methods have achieved significant performance improvements with minimal learning costs by transferring the general knowledge of natural image foundation models to the field of remote sensing. RSPrompter [6] introduces an innovative prompt learning method that incorporates the segmentation knowledge of the SAM foundation model, exhibiting remarkable performance on a wide range of data from different remote sensors. Recently, Mamba is more likely to be an alternative for the core component of foundation models [55], [56]. We have developed a general building understanding model and enhanced the model's generalization capability by amassing a substantial quantity of remote sensing building training data across various resolutions and geographical environments in this article.

D. Summary

Current interpretations of buildings in remote sensing imagery are in their infancy, primarily focusing on specific methodologies, datasets, and application scenarios. These approaches often treat the tasks of building extraction and CD as separate entities. In building extraction, the emphasis is primarily on the exploration of multiscale building features, precise segmentation of building edges, and refinement of occluded buildings [57]. Conversely, CD revolves around the extraction of more refined features, efficient temporal feature fusion, and the challenge of scarce positive change samples [58], [59].

Fundamentally, both building extraction and CD should align in the low-level semantic space, given that their research subjects are identical—buildings. The divergence lies in the focus: the former concentrates on the existence of buildings, while the latter highlights the changes over time. Our objective is to investigate the complementary between these two tasks within a unified framework, advancing from the perspective of the foundation model, and thereby liberating ourselves from the complexities of subbranch and submodule design. Specifically, we adhere to the widely successful pathway by employing a cumbersome encoder and a lightweight decoder.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

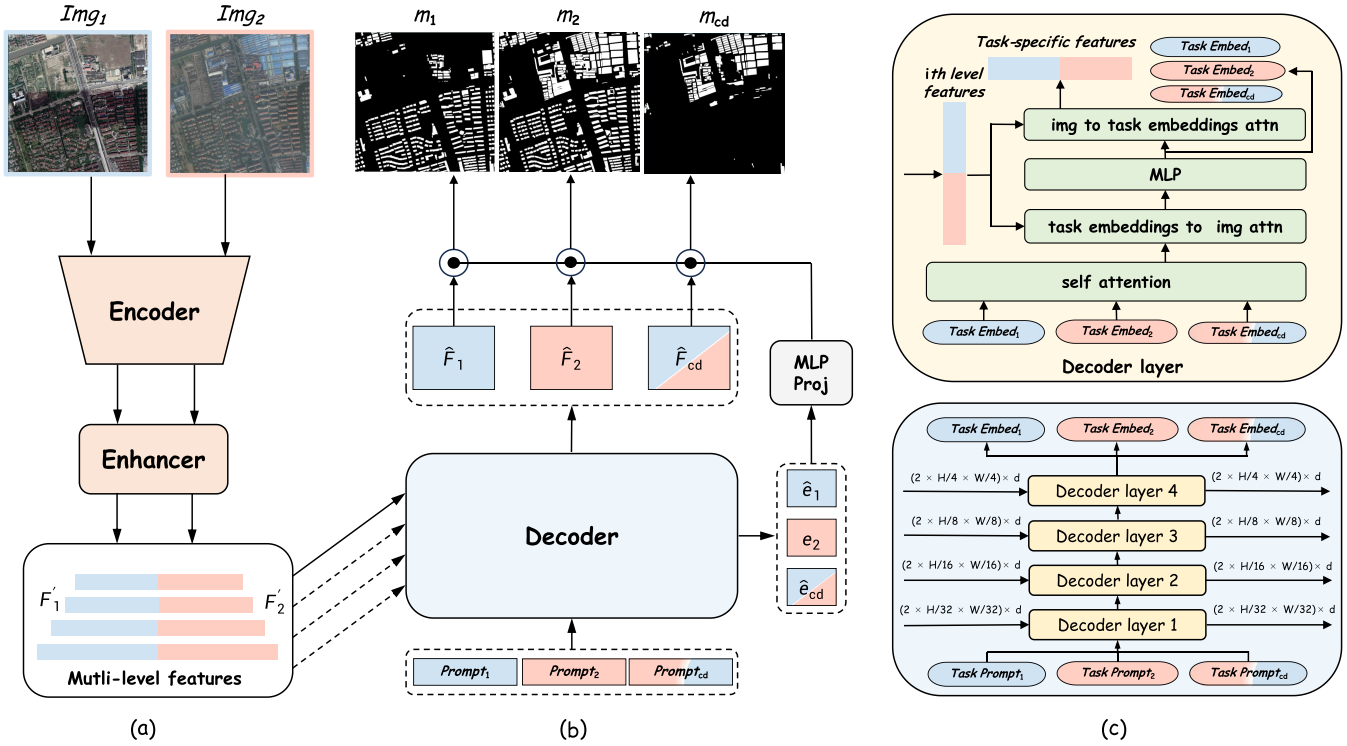


Fig. 2. Overview structure of the RSBuilding, a model for processing dual-temporal images to obtain individual building masks and building changes. The model primarily consists of four parts: an encoder for extracting dual-temporal robust features, an enhancer for enriching multiscale information, a decoder for conducting spatiotemporal information interaction and querying corresponding semantic mask features based on task clues, and a segmentation head for generating the final segmentation results by leveraging Einstein summation. (a) Backbone and enhancer. (b) Decoder and head. (c) Decoder architecture.

Additionally, we amass and restructure a substantial volume of annotated building data in various forms, and conduct comprehensive training on these data. This enables the cross-scene and cross-scale understanding of buildings within a unified framework. Our research will provide valuable insights into the evolution of foundation visual models in remote sensing.

III. Methodology

In this section, we will present an exhaustive introduction to RSBuilding, a general visual foundation model specifically designed for the interpretation of buildings in remote sensing imagery. The discussion will encompass a comprehensive overview of the model, an in-depth explanation of each constituent component, and an examination of the implemented federated training strategy.

A. Problem Setting

Consider a training dataset, denoted as $\mathcal{D}_{\text{train}} = \{(x_1, y_1), \ldots, (x_N, y_N)\}$. Each x_i represents a pair of images, specifically $x_i = \{\mathcal{I}_i^1, \mathcal{I}_i^2 \in \mathbb{R}^{H\times W\times 3}\}$. Correspondingly, $y_i = \{m_i^1, m_i^2, m_i^{\text{cd}}\}$ signifies the ground-truth annotations associated with x_i. Here, $m_i^1 and m_i^2 \in \mathbb{R}^{H\times W}$ are the building segmentation annotations for $\mathcal{I}_i^1$ and $\mathcal{I}_i^2$, respectively. Additionally, $m_i^{\text{cd}} \in \mathbb{R}^{H\times W}$ denotes the change mask between $\mathcal{I}_i^1$ and $\mathcal{I}_i^2$. It is worth noting that the validity of the building change segmentation mask m_i^{cd} hinges on the alignment of geographical spaces $\mathcal{I}_i^1$ and $\mathcal{I}_i^2$.

Our goal is to design and train a foundation model capable of processing any pair of images from a given test set ($x_k \sim \mathcal{D}_{\text{test}}$), concurrently generating the segmentation masks for buildings and the mask indicating changes of buildings over different periods

$$\{\hat{m}_k^1, \hat{m}_k^2, \hat{m}_k^{\text{cd}}\} = \Phi_{\text{head}} \circ \Phi_{\text{dec}} \circ \Phi_{\text{enhancer}} \circ \Phi_{\text{enc}}(x_k)$$

where a pair of images undergoes a sequential processing pipeline, comprising a robust encoder, a multilevel feature enhancer and decoder, and a streamlined segmentation head, to produce three binary segmentation masks. The overall architecture is illustrated in Fig. 2. For the sake of clarity in the ensuing discussion of the proposed model, the subscript k will be omitted.

B. Encoder

In this section, we introduce the architecture of the encoder embedded within the RSBuilding model. The encoder, a critical component endowed with a substantial quantity of learnable parameters, facilitates the derivation of robust visual features. Our approach adopts a paradigm wherein dual-temporal general building features are initially extracted independently. Subsequently, we query and express the effective features through the decoder. The encoding process is outlined as follows:

$$\begin{aligned}\{F_1^i\} &= \Phi_{\text{enc}}(\mathcal{I}_1)\\ \{F_2^i\} &= \Phi_{\text{enc}}(\mathcal{I}_2)\end{aligned} \tag{1}$$

where $\{F_1^i\}$ and $\{F_2^i\}$ are the respective sets of features extracted from $\mathcal{I}_1$ and $\mathcal{I}_2$. The superscript i represents the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

ith level of the feature. We employ two prevalent foundation architectures based on the transformer to establish the encoder; specifically, the vision transformer (ViT) [60] and the swin transformer (Swin) [61].

1) ViT-Based Siamese Encoder: ViT [60] divides the input image into non-overlapping patches of a fixed size. Each patch is then flattened and linearly projected into a fixed-length embedding. The algorithm employs stacked transformer encoders to model the interrelationships among these embeddings, thereby capturing the global semantic information inherent in the image. In our article, we leverage the backbone of the pre-trained SAM [19] foundation segmentation model and introduce variants with varying parameter quantities. Specifically, we present ViT-B with 12 layers and ViT-L with 24 layers. The encoder produces a single-scale feature map, that is, $\{F_1^i\} = \{F_1^1 \in \mathbb{R}^{(H/2^4)\times(W/2^4)\times d}\}$, where d represents the channel dimension of the feature map.

2) Swin-Based Siamese Encoder: The swin transformer [61] constrains the computation of attention to a specific window via a sliding window operation. This approach not only incorporates the locality inherent in CNN but also optimizes the use of computational resources. Different from the ViT, the Swin generates multiscale feature representations through a hierarchically structured design, that is, $\{F_1^i \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times d^i}\}, i \in \{1, 2, 3, 4\}$.

C. Multilevel Feature Enhancer

To tackle the complexities associated with multiresolution remote sensing imagery and objects of varying scales, we introduce a feature enhancer to generate semantic features across multiple scales. The inputs are derived from the outputs of various architectural encoders, ultimately resulting in multilevel features of different sizes but with consistent channel dimensions. The process can be expressed with the following formula:

$$\left\{F'^i_1\right\} = \Phi_{\text{enhancer}}\left(\left\{F_1^i\right\}\right) \tag{2}$$

where we only present the single-temporal image feature process for simplicity. The set $\{F_1^i\}$ represents the feature maps derived from the encoder of $\mathcal{I}_1$. $\{F'^i_1 \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times d}\}, i \in \{1, 2, 3, 4\}$, denotes the multilevel features. Specifically, we employ straightforward max-pooling down-sampling layers and transposed convolutional upsampling layers to generate multilevel feature maps for the ViT encoder. For the Swin encoder, we leverage a standard feature pyramid network (FPN) [62] layer. The reason for using different enhancers is that Swin encoder inherently has multiscale feature maps, while ViT encoder only has single-scale feature maps. The purpose of the multiscale enhancers is to create more targeted and rich feature information after the preliminary interaction of semantic and fine-grained features, allowing the decoder to perform better during feature extraction.

D. Decoder

To address the challenges of unifying building extraction and CD within a framework, we have developed an efficient multitask decoder that leverages the attention mechanism. More specifically, we employ three learnable query embeddings to aggregate semantic features pertinent to different tasks. These three query embeddings, denoted as e_1, e_2, and e_{cd}, correspond to the building extraction tasks for $\mathcal{I}_1$ and $\mathcal{I}_2$, and the building CD task between them, respectively. These query embeddings interact with the dual-temporal image features via a cross-attention mechanism, thereby modeling the spatiotemporal relationship and deriving task-specific feature representations. The decoder processes the dual-temporal features at various levels through a series of mixed cross-attention layers. The process can be recursively written as follows:

$$\begin{aligned}
E^0 &= \Phi_{\text{Cat}}(e_1, e_2, e_{\text{cd}}) \\
T_1^i &= F'^i_1 + \text{PE} + \text{LE}_1 \\
T_2^i &= F'^i_2 + \text{PE} + \text{LE}_2 \\
T^i &= \Phi_{\text{Cat}}\left(\Phi_{\text{Flatten}}\left(T_1^i\right), \Phi_{\text{Flatten}}\left(T_2^i\right)\right) \\
E^i &= \Phi_{\text{S-attn}}\left(E^{i-1}\right) \\
E^i &= \Phi_{\text{C-attn}}\left(E^i, T^i\right) \\
E^i &= \Phi_{\text{mlp-proj}}\left(E^i\right) \\
T^i &= \Phi_{\text{C-attn}}\left(T^i, E^i\right)
\end{aligned} \tag{3}$$

where E^0 represents the task-specific embedding at the zeroth level. The symbol Φ_{Cat} denotes the operation of vector concatenation along the sequence dimension. PE refers to a pre-set learnable positional encoding map. The temporal encodings of $\mathcal{I}_1$ and $\mathcal{I}_2$ are represented by LE_1 and LE_2, respectively. The operation Φ_{Flatten} signifies the process of flattening a 2-D feature map into a 1-D sequential vector. $T^i \in \mathbb{R}^{(2\times(H/2^{i+1})\times(W/2^{i+1}))\times d}$ denotes the ith level concatenated dual-temporal image features. The standard multihead self-attention layer is represented by $\Phi_{\text{S-attn}}$, while the multihead cross-attention layer is denoted by $\Phi_{\text{C-attn}}$. $\Phi_{\text{mlp-proj}}$ refers to a linear projection layer.

We follow the operation outlined below to obtain task-specific dense feature representations, which are used to interpret the corresponding masks

$$\begin{aligned}
\hat{F}_1^i, \hat{F}_2^i &= \Phi_{\text{Reshape}}\left(\Phi_{\text{Split}}\left(T^i\right)\right) \\
\hat{F}_1^i &= \Phi_{\text{Up}}\left(\hat{F}_1^i\right) \\
\hat{F}_1 &= \Phi_{\text{CLR}}\left(\Phi_{\text{Cat}}\left(\left\{\hat{F}_1^i\right\}\right)\right) \\
\hat{F}_{\text{cd}} &= \hat{F}_1 - \hat{F}_2
\end{aligned} \tag{4}$$

where T^i is split and reshaped to reestablish the dual-temporal 2-D feature maps, denoted as $\hat{F}_1^i \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times d}$ and $\hat{F}_2^i \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times d}$, where i is an element of the set $\{1, 2, 3, 4\}$. The operator Φ_{Up} signifies the process of upsampling feature maps to a unified scale of $(H/4) \times (W/4)$, achieved through bi-linear interpolation. The operator Φ_{Cat} is used to represent the concatenation of feature maps along the channel dimension. The term Φ_{CLR} denotes a Conv-LayerNorm-ReLU layer with a kernel size of 1×1 to reduce the channel dimension to 64. This is followed by another Conv-LayerNorm-ReLU layer with a kernel size of 3×3 to integrate the spatial information. $\hat{F}_2$ is generated in the same manner as $\hat{F}_1$. $\hat{F}_{\text{cd}} \in \mathbb{R}^{(H/4)\times(W/4)\times 64}$ denotes the changed dense features between the dual-temporal images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

E. Segmentation Head

To derive the interpretive results for various tasks, we employ task-specific embedding vectors to filter the dense feature, that is, Einstein summation, illustrated as follows:

$$\begin{aligned} \hat{E} &= \Phi_{\text{e-proj}}\left(E^4\right) \\ \hat{e}_1, \hat{e}_2, \hat{e}_{\text{cd}} &= \hat{E} \\ \hat{m}_1 &= \Phi_{\text{e-sum}}\left(\hat{F}_1, \hat{e}_1\right) \end{aligned} \tag{5}$$

where $\Phi_{\text{e-proj}}$ denotes a linear projection layer to diminish the dimension of the final-layer task-specific embedding. The segmentation mask, $\hat{m}_1$, is computed by $\Phi_{\text{e-sum}}$ through linear weighting $\hat{F}_1$ using $\hat{e}_1$. It should be noted that $\hat{m}_2$ and $\hat{m}_{\text{cd}}$ are derived in the same manner.

F. Training Strategy

The objective of our study is to address the tasks of building extraction and CD from the perspective of the foundation model, all within a unified framework, and validate the universality and complementarity of the intrinsic knowledge shared between the two tasks. Ideally, we aspire to possess an extensive collection of bi-temporal data pairs, $\mathcal{I}_1 and \mathcal{I}_2$, accompanied by corresponding labels for building segmentation and change, m_1, m_2, and m_{cd}. However, in practical scenarios, only the BANDON [63] and xView2 [64] datasets provide such comprehensive annotations. Unfortunately, the volume of the datasets is limited and falls short of the requirements for high-level generalization. To mitigate this limitation, we have curated and structured a substantial quantity of datasets for building extraction and CD, collectively referred to as the RSBuilding dataset. It is noteworthy that the data may solely encompass annotation information pertinent to a single task.

We employ binary cross-entropy (CE) loss for the supervised training. The overall loss function is as follows:

$$\begin{aligned} \mathcal{L} = {} & \alpha_1 \text{CE}\left(\hat{m}_1, m_1\right) + \alpha_2 \text{CE}\left(\hat{m}_2, m_2\right) \\ & + \alpha_{\text{cd}} \text{CE}\left(\hat{m}_{\text{cd}}, m_{\text{cd}}\right) + \alpha_{\text{p-cd}} \text{CE}\left(\left|\hat{m}_1 - \hat{m}_2\right|, m_{\text{cd}}\right) \end{aligned} \tag{6}$$

where $|\hat{m}_1 - -\hat{m}_2|$ denotes the pseudo CD prediction. To accommodate a diverse need for tasks and training data within a unified training loss framework, we have the following conventions for $\alpha_1, \alpha_2, \alpha_{\text{cd}}$, and $\alpha_{\text{p-cd}}$.

1) For the building extraction task, which includes building segmentation annotations, we create a duplicate of $\mathcal{I}_1$, denoted as $\mathcal{I}_2$, and apply random photometric distortion. The parameters are set as follows: $\alpha_1 = 1, \alpha_2 = 1, \alpha_{\text{cd}} = 0$ (to mitigate the introduction of negative samples), $\alpha_{\text{p-cd}} = 0.1$, and $m_{\text{cd}} = 0$.
2) For the CD task, which solely includes change masks, the parameters are adjusted to $\alpha_1 = 0, \alpha_2 = 0, \alpha_{\text{cd}} = 1$, and $\alpha_{\text{p-cd}} = 0.1$ (to minimize the impact of geometric location errors in bi-temporal images).
3) For the CD task that includes comprehensive annotations, the parameters are $\alpha_1 = 1, \alpha_2 = 1, \alpha_{\text{cd}} = 1$, and $\alpha_{\text{p-cd}} = 0.1$.

TABLE I
SUMMARY OF DATASET STATISTICS

Dataset	Train	Eval.	Task	Images
WHU [65]	✓	✓	Seg	8k
Massachusetts[66]	✓	✗	Seg	2k
INRIA [67]	✓	✓	Seg	12k
Map Challenge [68]	✓	✗	Seg	90k
Tanzania Challenge [69]	✓	✗	Seg	6k
EGY-BCD [70]	✓	✗	CD	6k
MSBC [71]	✓	✗	CD	4k
MSOSCD [71]	✓	✗	CD	5k
LEVIR-CD [9]	✓	✓	CD	3k
S2Looking [72]	✓	✓	CD	20k
WHU-CD [65]	✓	✗	CD	2k
BANDON [63]	✓	✓	Seg & CD	47k
xView2 [64]	✓	✗	Seg & CD	40k

IV. Experimental Results and Analyses

A. Experimental Datasets and Settings

In this article, we present the RSBuilding dataset, an extensive and large-scale dataset meticulously curated to train foundation models for various building understanding tasks. This dataset includes annotated data specifically designed for tasks such as building extraction and building CD. The entire dataset can be primarily divided into three categories.

1) *Building Segmentation Masks for Single-Temporal Images*: WHU building dataset [65], Massachusetts building dataset [66], INRIA building dataset [67], Map Challenge [68], and the 2018 Open AI Tanzania Building Footprint Segmentation Challenge [69].
2) *Building Change Masks for Dual-Temporal Images*: EGY-BCD [70], MSBC [71], MSOSCD [71], LEVIR-CD [9], S2Looking [72], and WHU-CD [65].
3) *Building Segmentation and Change Masks for Dual-Temporal Images*: BANDON [63] and xView2 [64].

An overview of the dataset used for training and testing is provided in Table I. All the data are cropped to a unified size of 512 × 512. We have selectively gathered partial data from both the Map Challenge and Tanzania Challenge datasets. To demonstrate the efficacy of the proposed method, we conducted exhaustive experimental validation using two publicly available building extraction datasets (WHU and INRIA), two CD datasets (LEVIR-CD and S2Looking), and a dual-label dataset (BANDON). It is imperative to note that the test data relevant to performance validation have been excluded from the RSBuilding training dataset.

1) WHU [65]: The WHU building dataset is composed of two sub-datasets: one for satellite images and another for aerial images. Our study utilizes the aerial image sub-dataset, which encompasses 8189 images. These images are further divided into 4736 for training, 1036 for validation, and 2416 for testing. Each image has a spatial resolution of 0.3 m. The entire aerial image sub-dataset includes approximately 22 000 buildings, spanning an area exceeding 450 km^2.

2) INRIA [67]: The INRIA building dataset comprises 360 images from five distinct cities: Austin, Chicago, Kitsap, Tyrol, and Vienna. Following the official recommendation, we selected the first to fifth images from each city for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

validation, with the remaining images allocated for training. The original 5000 × 5000 pixel images were cropped to 512 × 512 pixels for both training and validation purposes.

3) LEVIR-CD [9]: The LEVIR-CD dataset includes 637 pairs of bi-temporal remote sensing images, along with over 31 333 instances of change annotation. These images were sourced from Google Earth. Each image pair measures 1024 × 1024 pixels, with a resolution of 0.5 m/pixel. Following the official guidelines, we divided the dataset into 445, 64, and 128 image pairs for training, validation, and testing, respectively.

4) S2Looking [72]: The S2Looking dataset contains 5000 image pairs, each measuring 1024 × 1024 pixels. These pairs are divided into a ratio of 7:1:2 for training, evaluation, and testing, respectively. The dataset includes over 65 920 instances of change annotation. The images, collected by optical satellites around the world, span a period of one to three years and have a resolution ranging from 0.5 to 0.8 m/pixel. For training and validation, these images were cropped to a size of 512 × 512 pixels.

5) BANDON [63]: The BANDON dataset is designed to assess the generality and task complementarity of the proposed method. It consists of aerial images with a spatial resolution of 0.6 m. The images are sourced from various platforms, including Google Earth, Microsoft Virtual Earth, and ArcGIS. The dataset includes geographic images from six representative cities in China: Beijing, Shanghai, Wuhan, Shenzhen, Hong Kong, and Jinan. BANDON comprises 1689 training image pairs, 202 validation image pairs, and 392 testing image pairs, each measuring 2048 × 2048 pixels. Each image is accompanied by bi-temporal building segmentation labels and change labels.

B. Evaluation Protocol and Metrics

We conducted an extensive and comprehensive evaluation of the proposed model using widely accepted metrics in segmentation tasks, including intersection over union (IoU), precision (P), recall (R), and $F1$ score ($F1$). The definitions of these four evaluation metrics are as follows:

$$\begin{aligned} \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \\ \text{P} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{R} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1} &= \frac{2\times\text{P}\times\text{R}}{\text{P}+\text{R}} \end{aligned} \tag{7}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

C. Implementation Details

Our research is primarily concentrated on the development of a streamlined foundation model architecture, aiming to enhance generalization and versatility in the semantic interpretation of buildings for remote sensing imagery. We approach this problem from the perspective of the foundation model. Unless stated otherwise, our experiments involve multitask federated training utilizing the amassed RSBuilding dataset, as opposed to being confined to a single dataset.

1) Architecture Details: To accommodate heterogeneous tasks within a unified framework and fulfill the demands of both single-image building extraction and dual-image CD tasks, we adopt a dual-image input scheme as a compromise. Each image is processed through an encoder and enhancer to extract high-level semantic features. These features are subsequently fused and queried by a decoder to produce three mask maps. For the encoder, we employ either the SA-1B [19] pre-trained ViT [60] or the ImageNet 22k [73] pre-trained Swin [61]. The position encoding of the encoder is adjusted via interpolation to fit a fixed input size of 512 × 512.

2) Training Details: During the training phase, we adhere to a fixed input image size of 512 × 512 and apply data augmentation techniques such as random cropping, flipping, photometric distortion, mosaic, and temporal exchange. In the testing phase, the images to be validated are cropped to a size of 512 × 512, with no additional validation-time augmentation methods applied. We employ binary CE loss for supervised training and utilize the AdamW optimizer with an initial learning rate of $1e^{-4}$, a weight decay of 0.05, and a poly scheduler to decay the learning rate. The batch size for training is set at 16. The foundation model underwent 400k training iterations, and the experiment was conducted on eight NVIDIA A100 GPUs. The algorithm was developed based on the OpenCD [42] and MMSegmentation [74] platform, with PyTorch serving as the underlying deep learning framework.

D. Comparison With the State-of-the-Art

We evaluate the effectiveness of our proposed method, RSBuilding, by comparing it with several state-of-the-art methods for building extraction and CD. The methods chosen for comparison encompass a wide range of approaches, including: general semantic segmentation methods such as UNet [35], Deeplab-v3 [75], and SegFormer [37]. Remote sensing image semantic segmentation and building extraction methods, including MA-FCN [34], DSNet [77], MSNet [78], BOMSNet [79], LCS [80], BuildFormer [12], BCTNet [81], and FD-Net [82]. CD methods such as FC-Siam-Conc, FC-Siam-Diff [86], SNUNet-C32 [83], BiT [3], ChangeFormer [17], TinyCD [84], HANet [85], and ChangerEx [42]. The performance metrics presented for these methods are based on either their officially published results or our re-implementation using PyTorch, adhering closely to the specifications provided in their respective publications.

1) Quantitative and Qualitative Comparisons for Building Extraction: In this section, we will demonstrate the performance of RSBuilding in building extraction tasks using the Swin-B as the backbone model, with quantitative metrics compared as shown in Table II. The proposed method performs optimally on both the WHU and INRIA datasets, with an IoU of 92.15% and 82.68%, respectively. When compared with other contemporary methods, our approach continues to exhibit significant advantages. This verified the outstanding performance of the proposed method in building extraction tasks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF BUILDING EXTRACTION PERFORMANCE USING VARIOUS METHODS ON WHU AND INRIA TEST SETS

Method	Year	WHU (%)				INRIA (%)			
		P	R	F1	IoU	P	R	F1	IoU
UNet [35]	2015	94.52	94.77	94.52	89.60	87.21	86.09	86.65	76.44
Deeplab-v3 [75]	2017	94.80	93.94	94.36	89.33	90.30	87.74	89.00	80.18
Segformer [37]	2021	94.72	94.42	94.57	89.70	88.97	87.08	88.01	78.59
HRNet [76]	2019	91.69	92.85	92.27	85.64	86.56	84.92	85.73	75.03
MA-FCN [34]	2019	95.20	95.10	95.15	90.70	89.82	87.58	88.68	79.67
DSNet [77]	2020	94.05	94.91	94.48	89.54	90.32	88.73	89.52	81.02
MSNet [78]	2022	94.83	93.12	93.96	89.07	-	-	-	-
BOMSNet [79]	2022	95.14	94.50	94.80	90.15	87.93	87.58	87.75	78.18
LCS [80]	2022	95.38	94.86	95.12	90.71	89.58	86.77	88.15	78.82
BuildFormer [12]	2022	95.15	95.14	95.14	90.73	90.65	88.78	89.71	81.24
BCTNet [81]	2023	95.47	95.27	95.37	91.15	-	-	-	-
FD-Net [82]	2023	95.27	95.46	95.36	91.14	-	-	-	-
RSBuilding (Ours)	-	**95.93**	**95.82**	**95.88**	**92.15**	**91.40**	**89.65**	**90.52**	**82.68**

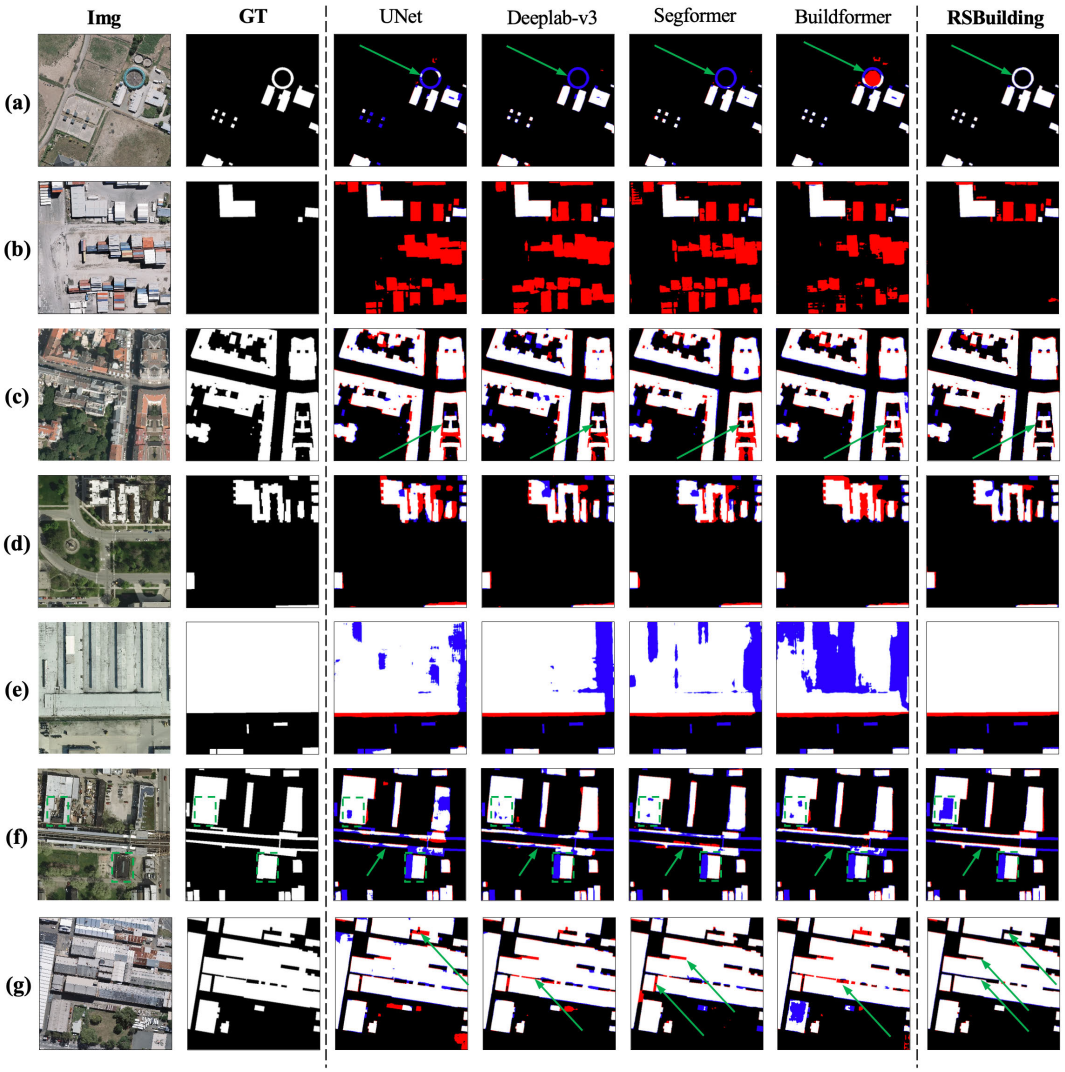


Fig. 3. Visual comparisons of the proposed method with other state-of-the-art methods for building extraction. Red means FPs, while blue denotes FNs. Samples are all from the WHU and INRIA building test sets.

Accurately representing complex building shapes, semantics, and edges is crucial for effective building extraction. Fig. 3 visually illustrates the performance of our proposed method compared with other advanced methods. Specifically, Fig. 3(a) highlights RSBuilding's proficiency in segmenting rare-shaped buildings and small-scale targets. Fig. 3(b)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF CD PERFORMANCE USING VARIOUS METHODS ON LEVIR-CD AND S2LOOKING TEST SETS

Method	Year	LEVIR-CD (%)				S2Looking (%)			
		P	R	F1	IoU	P	R	F1	IoU
FC-Siam-Conc[41]	2018	87.09	90.10	88.57	79.48	**73.59**	30.09	42.72	27.16
FC-Siam-Diff[41]	2018	88.47	89.55	89.00	80.19	72.39	34.53	43.95	28.16
SNUNet-C32 [83]	2021	**93.45**	90.54	91.97	85.14	70.53	58.21	63.77	46.81
BiT [3]	2021	93.11	90.21	91.64	84.56	72.96	56.38	63.61	46.63
Changeformer [17]	2022	93.03	89.66	91.32	84.02	71.41	61.42	66.04	49.30
TinyCD [84]	2023	92.39	92.84	90.51	84.53	71.07	58.48	64.16	47.24
TinyCDv2-L [84]	2023	92.40	90.91	91.64	84.58	72.01	54.08	61.77	44.69
HANet [85]	2023	93.07	89.42	91.21	83.84	63.51	56.36	59.72	42.57
ChangerEx-RS50 [42]	2023	90.48	**93.09**	91.77	84.79	71.96	61.88	66.54	49.86
RSBuilding (Ours)	-	93.39	91.80	**92.59**	**86.19**	73.14	**64.97**	**68.81**	**52.46**

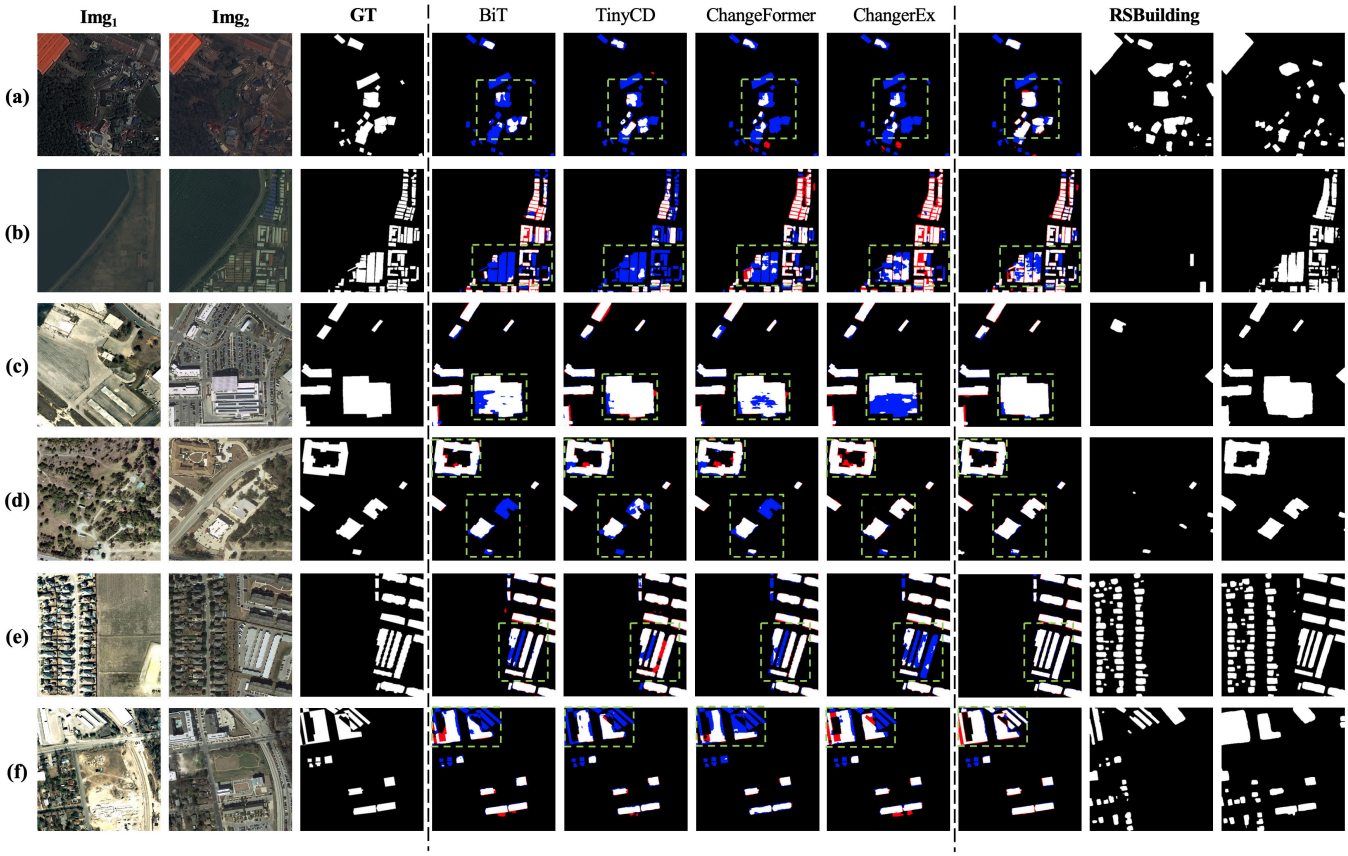


Fig. 4. Visual comparisons of the proposed method with other state-of-the-art methods for CD. Red means FPs, while blue denotes FNs. Samples are all from the LEVIR-CD and S2Looking test sets. The final two columns depict building extraction results from the pre-temporal and post-temporal images.

showcases its precise semantic expression, preventing misclassification of densely arranged containers as buildings. Fig. 3(c) and (d) demonstrates its robust discriminative power in distinguishing complex and confusing remote sensing backgrounds. Fig. 3(e) exemplifies its global understanding capability, enabling it to handle buildings at different resolutions, including large structures spanning the entire image. Fig. 3(f) and (g) underscores its precision edge expression abilities. Overall, RSBuilding exhibits strong cross-scene generalization, adapting effectively to various complex scenarios and multiscale transformations.

2) Quantitative and Qualitative Comparisons for CD: The comparative results of CD tasks between RSBuilding and other methods are delineated in Table III. We employ ViT-L as the foundation backbone of RSBuilding. Notably, RSBuilding outperforms other comparative methods on the LEVIR-CD and S2Looking datasets with an IoU of 86.19% and 52.46%, respectively. It exhibits robust proficiency in discerning temporal changes within remote sensing imagery.

The effectiveness of building CD hinges on the ability to discern meaningful change amidst irrelevant variations. Fig. 4 highlights the exceptional performance of RSBuilding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF BUILDING EXTRACTION AND CD PERFORMANCE USING VARIOUS METHODS ON THE BANDON TEST SET

Method	Year	Building Extraction (%)				Change Detection (%)			
		P	R	F1	IoU	P	R	F1	IoU
UNet [35]	2015	86.60	86.93	86.76	77.19	-	-	-	-
Deeplab-v3 [75]	2017	86.48	88.74	87.66	78.53	-	-	-	-
Segformer [37]	2021	86.81	87.79	87.30	78.05	-	-	-	-
Buildformer [12]	2022	82.26	88.22	87.74	78.75	-	-	-	-
FC-Siam-Diff [41]	2018	-	-	-	-	72.80	31.84	44.31	28.46
SNUNet-C32 [83]	2021	-	-	-	-	70.43	56.97	62.99	45.97
BiT [3]	2021	-	-	-	-	71.16	65.64	68.29	51.85
Changeformer [17]	2022	-	-	-	-	74.48	66.24	70.12	53.98
TinyCD [84]	2023	-	-	-	-	73.01	62.71	67.47	50.91
ChangerEx [42]	2022	-	-	-	-	72.69	68.56	70.56	54.52
RSBuilding (Ours)	-	**88.82**	**89.64**	**89.23**	**80.55**	**75.55**	**72.44**	**73.96**	**58.68**

Specifically, Fig. 4(a) demonstrates RSBuilding's capability in managing complex environments and processing degraded images. Fig. 4(b) and (c) highlights its prowess in describing changes across various scales of targets. Fig. 4(d) demonstrates a more precise extraction of change edges. Fig. 4(e) and (f) exhibits an impressive ability to perceive high-level semantic information related to changes. This enables more accurate change delineation compared to alternative methods that struggle to define changes and tend to produce more FPs. Moreover, RSBuilding's robust generalization is commendable. Even without single-temporal building segmentation mask supervision annotations, it can accurately extract building masks for each corresponding temporal image; thanks to its multitask federated training approach.

3) Quantitative and Qualitative Comparisons for Both Tasks: To evaluate the performance of both building extraction and CD tasks, we conducted an extensive evaluation using the BANDON dataset with ViT-L as the backbone, which provides annotations for both tasks. The experimental results indicate that RSBuilding achieves impressive IoU scores of 80.55% and 58.68% for the respective tasks, as illustrated in Table IV. Significantly, these metrics outperform those achieved by task-specific training methods. The experimental observations not only emphasize the complementary relationship between the two tasks but also underscore the multitask representation capabilities inherent in the proposed framework.

The segmentation results of the proposed method for two distinct tasks are illustrated in Fig. 5. We can make the following observations: Fig. 5(a) displays a large building located in the lower-right corner. This building exhibits noticeable differences between time periods A and B, primarily attributed to decorative modifications rather than any construction or demolition activities. Consequently, we argue that these alterations should not be considered as changes. Unlike other methods, RSBuilding delves deeper into the fundamental "essence" of changes, effectively minimizing irrelevant variations. Fig. 5(b) exemplifies the dual capabilities of our method: building extraction and CD. Specifically, RSBuilding can accurately delineate edges and represent complex architectural forms. Consequently, the predictions closely correspond with the actual shapes of the buildings, thereby enhancing the precision. Fig. 5(c) presents densely arranged yet noncontiguous buildings. The existence of shadows between building intervals poses a significant challenge in distinguishing individual structures. In this context, RSBuilding not only provides more precise edge predictions but also reduces interference from shadows and other confounding factors during the process of identifying individual buildings.

E. Ablation Study

In this section, we aim to validate the efficacy of the proposed components and strategies by conducting a series of ablation experiments. Unless otherwise specified, we employ the ViT-L, pre-trained on the SA-1B dataset, as the backbone of the proposed model.

1) Effects of Different Encoder Architectures: The parameter size of the visual foundation model's image encoder can significantly affect the inference speed. An encoder architecture with a robust feature representation capability can yield better performance within a relatively low inference latency. In our study, we conduct experiments utilizing the SA-1B pre-trained ViT and the ImageNet pre-trained swin transformer as the encoders, respectively. The swin transformer exhibits advantages in modeling data with varying scales, while the ViT is more concise, demonstrating a strong inductive bias capability. Specifically, we employ Swin-T, Swin-B, ViT-B, and ViT-L for comparative experiments.

Table V delineates the performance of various encoder architectures in building interpretation. The experimental results indicate that when all models are base versions with similar parameter quantities ($\sim$ 80M), Swin exhibits certain advantages in modeling buildings and shows better performance on building segmentation. As the model size is further expanded to the large version, there is a corresponding improvement in performance indicators. On the whole, the architectures of Swin and ViT exert a marginal influence on the performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

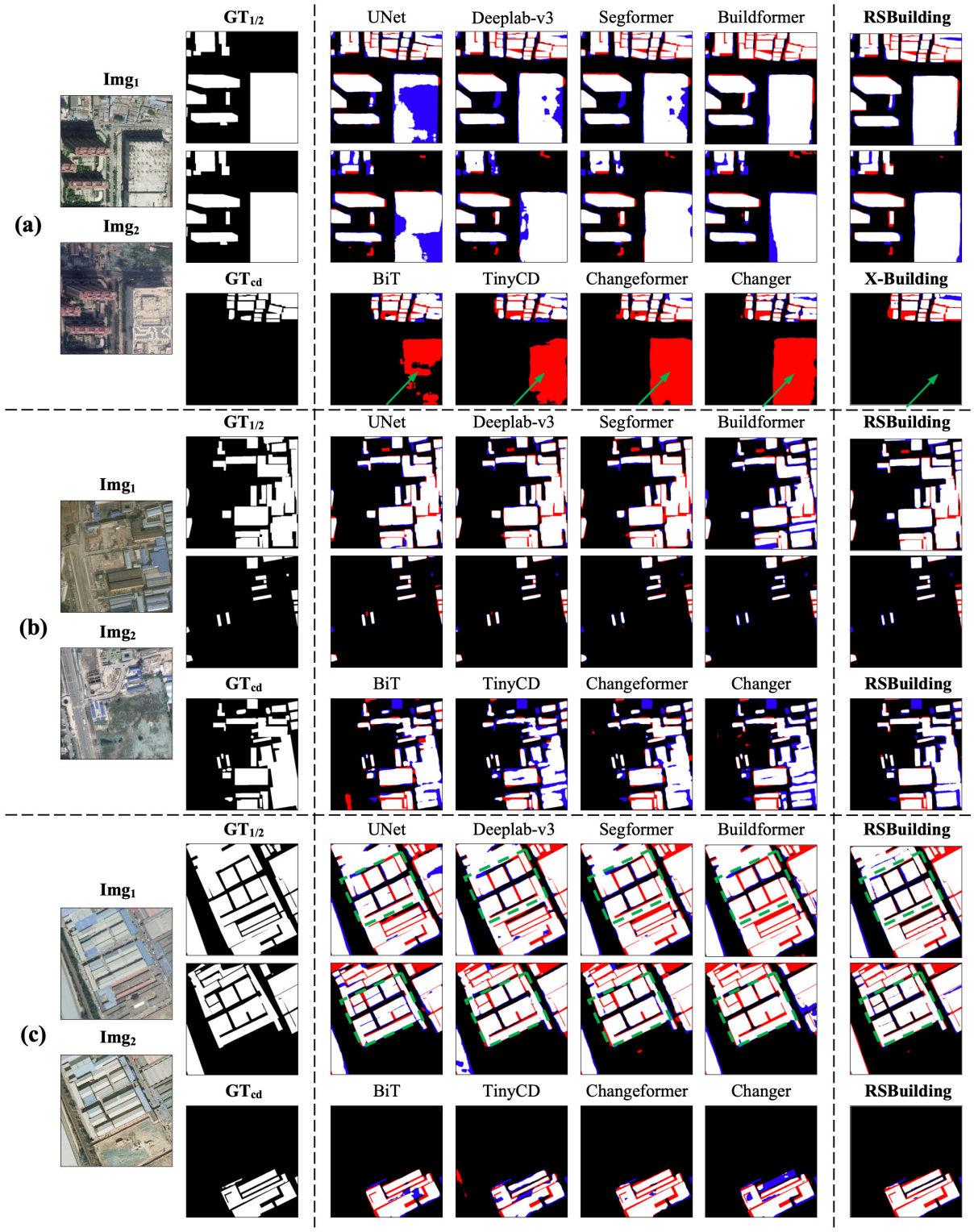


Fig. 5. Visual comparisons of the proposed method with other state-of-the-art methods for building extraction and CD. Red means FPs, while blue denotes FNs. Samples are all from the BANDON test sets.

2) Effects of Model Components: To achieve a general model for multitasking and to extract more generalized feature representations, we have introduced a multiscale feature enhancer, a dual-path cross-attention decoder, and a multilevel decoding mechanism into our approach. To validate the performance improvement attributed to these specifically designed sub-modules, we executed a series of ablation experiments. The results are illustrated in Table VI. Our observations are as follows.

1) Considering the diverse resolution of remote sensing images and the variable scale of building targets, we incorporated a multilevel feature enhancer into the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

TABLE V

EFFECTS OF DIFFERENT ENCODER ARCHITECTURES. PERFORMANCES ARE VALIDATED ON THE WHU, INRIA, LEVIR-CD, S2LOOKING, AND BANDON TEST SETS. BANDON-BX MEANS BUILDING EXTRACTION PERFORMANCE, WHILE BANDON-CD DENOTES CD PERFORMANCE. PARA. REPRESENTS THE PARAMETER QUANTITY

Architecture	Para. (M)	WHU		INRIA		LEVIR-CD		S2Looking		BANDON-BX		BANDON-CD	
		IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
Swin-T	28	91.25	95.42	81.95	90.08	85.06	91.93	50.68	67.26	79.37	88.50	57.26	72.82
Swin-B	88	92.15	95.88	82.68	90.52	85.57	92.23	51.59	68.01	80.01	88.90	58.21	73.59
ViT-B	89	90.74	95.15	82.03	90.13	85.76	92.34	50.76	67.34	80.07	88.93	58.06	73.47
ViT-L	308	91.36	95.42	82.59	90.46	86.19	92.59	52.46	68.81	80.55	89.23	58.68	73.96

TABLE VI

EFFECTS OF DIFFERENT MODEL COMPONENT DESIGNS (MULTISCALE FEATURE ENHANCER, DUAL-PATH CROSS-ATTENTION DECODER, AND MULTILEVEL DECODING MECHANISM). PERFORMANCES ARE VALIDATED ON THE WHU AND INRIA TEST SETS

enhancer	decoder	decoding	INRIA		LEVIR-CD	
			IoU	F1	IoU	F1
			81.31	89.62	84.82	91.76
✓			81.78	89.91	85.37	92.09
✓	✓		82.05	90.13	85.73	92.32
✓	✓	✓	**82.59**	**90.46**	**86.19**	**92.59**

TABLE VII

EFFECTS OF DIFFERENT PRE-TRAINING DATASETS. PERFORMANCES ARE VALIDATED ON THE INRIA AND LEVIR-CD TEST SETS. BX MEANS BUILDING EXTRACTION DATA, WHILE CD MEANS CD DATA

Data	INRIA		LEVIR-CD	
	IoU	F1	IoU	F1
-	81.87	89.99	85.33	92.08
BX	82.40	90.33	85.70	92.30
CD	82.19	90.21	85.97	92.46
BX & CD	**82.59**	**90.46**	**86.19**	**92.59**

TABLE VIII

EFFECTS OF THE PSEUDO CD LOSS, REFERRED TO AS AUX LOSS, ARE VALIDATED ON THE WHU AND INRIA TEST SETS

Aux Loss	INRIA		LEVIR-CD	
	IoU	F1	IoU	F1
-	81.87	89.99	85.33	92.08
✗	82.17	90.19	85.84	92.37
✓	**82.59**	**90.46**	**86.19**	**92.59**

model. The performance enhancements observed in the two tasks substantiate the effectiveness of this design.

2) Compared to the single-path cross-attention derived from the plain transformer, the dual-path cross-attention (which mutually constructs attention through preset task-prompt tokens and dual-image feature tokens) yielded an IoU increase for both building extraction and CD tasks. This implies that the bidirectional attention decoder can facilitate a more efficient interaction between image features and query embeddings.
3) Given the significance of integrating high-level and low-level semantics in decoding, we separately incorporated multilevel features with task-specific semantic tokens during the decoding process, and unified the resulting semantic features to the same scale for concatenation. The experimental results indicate that this design further contributes to performance enhancements. The ablation studies demonstrate the effectiveness of the components in our model.

3) Effects of Federated Training: We leveraged the amassed data from both building extraction and CD tasks for the multitask federated training. To evaluate the impact of each data segment on the model's performance, we employed three different dataset settings: the building extraction (BX) dataset, the CD dataset, and the foundation dataset (BX and CD), that is, all data for model pre-training. After pre-training on these data segments, we have individually fine-tuned the pre-trained model on two downstream validation datasets. The experimental results are presented in Table VII. As discernible from the table, utilizing either individual data segments or the entire dataset can all result in performance enhancements across both tasks. Furthermore, to assess the role of the pseudo CD loss in federated training, we presented the impact of utilizing the pseudo CD loss in Table VIII. It is evident that the pseudo CD loss brings about significant benefits, especially for the building extraction task. These insights not only corroborate the complementary nature of the two tasks but also validate the efficacy of the proposed architecture in synergistically handling two heterogeneous tasks and achieving performance enhancements.

4) Generalization Ability: We aim to propose a general and generalized model for the building interpretation. To validate the generalization capability of the proposed method across diverse scenes and datasets, we have conducted experiments on the WHU-EAST ASIA [65] and SECOND [87] datasets, following three validation settings: zero-shot generalization, linear probes, and full parameter fine-tuning. The results were simultaneously compared with other advanced mainstream methods. Notably, our RSBuilding dataset was not contaminated by these two datasets, that is, these data were not incorporated into the training data.

The WHU-EAST ASIA dataset [65] comprises six contiguous satellite images that cover an area of 860 km^2 in East Asia, with a spatial resolution of 0.45 m. This dataset is frequently employed to assess the generalization capability of models across different data sources. We conducted experiments using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
GENERALIZATION PERFORMANCE ON THE WHU-EAST ASIA (BUILDING EXTRACTION) AND SECOND (CD) DATASETS, BOTH OF WHICH HAVE BEEN ENTIRELY EXCLUDED FROM THE TRAINING SET. FOR COMPARATIVE PURPOSES, WE HAVE INCLUDED SOME STATE-OF-THE-ART METHODS THAT EMPLOY FULLY SUPERVISED TRAINING. THE FIRST SECTION OF THE TABLE PRESENTS THE PERFORMANCE FOR BUILDING EXTRACTION, WHILE THE SECOND DELINEATES THE PERFORMANCE FOR CD

Method	P	R	F1	IoU
UNet	84.73	80.28	82.44	70.13
Segformer	84.33	82.75	83.53	71.72
Deeplab-v3	85.21	82.65	83.91	72.28
Buildformer	85.98	82.14	84.01	72.44
Zero-shot	87.69	79.35	83.31	71.42
Linear Probes	85.69	82.85	84.25	72.78
Fine-tuning	**86.09**	**83.96**	**85.01**	**73.93**
BiT	80.42	73.42	76.76	62.28
TinyCD	79.38	71.71	75.35	60.45
Changeformer	81.71	74.85	78.13	64.11
Changer-Ex	81.80	76.22	78.90	65.17
Zero-shot	81.61	67.44	73.91	58.96
Linear Probes	82.78	78.36	80.51	67.38
Fine-tuning	**84.83**	**79.79**	**82.23**	**69.83**

the second portion of this data; specifically, 3136 images for training and 904 images for testing.

The SECOND [87] dataset is a multicategory semantic CD dataset that covers cities such as Hangzhou, Chengdu, and Shanghai. It contains 4662 pairs of 512 × 512 aerial images. Our focus was on changes in the building category, with other categories being treated as background. As only the training set of this data is available, we divided the 2969 image pairs in the training set into training and testing sets at a ratio of 8:2.

To thoroughly verify the generalization capability of the pre-trained foundation RSBuilding model, we conducted experiments in three settings: 1) zero-shot, where the foundation model trained on the RSBuilding dataset is directly used for the building understanding tasks; 2) linear probes, where the majority of the model parameters are frozen and only the segmentation head of the foundation model is fine-tuned; and 3) fine-tuning, where all parameters of the model are further trained following the pre-training and then fine-tuning paradigm.

The experimental results are presented in Table IX. As can be seen from the table, even within the zero-shot setting, the model's performance achieves a level comparable to other supervised training methods and even surpasses some mainstream methods in the building extraction task. Through linear probing and supervised fine-tuning, the performance is further enhanced, surpassing other methods in all metrics. These findings underscore that the general building interpretation model proposed in this article possesses robust generalization ability, and can compete with current advanced methods even without any fine-tuning. The reasons for the excellent generalization can be summarized as follows: first, using a foundation model as the core allows for leveraging advanced features from various images. Second, the training dataset's wide geographical, architectural, and environmental variety enhances the model's universal representation, aiding generalization. Finally, a cooperative learning loss function improves generalization by optimizing tasks with cross-domain data during training.

F. Discussion

In this article, we introduce a general multitask understanding framework for buildings based on foundation model. Within this framework, we propose a multitask federated training strategy that accommodates partially missing task labels, thereby validating the complementarity of building extraction and building detection tasks. The model is constructed on the widely used transformer structure, integrating the fundamental ViT and swin transformer structure as the large-scale visual encoder. During experiments, we verified the performance of both the base and large versions. The results indicated a minimal performance discrepancy between ViT and Swin, with an observed increase in performance correlating with the expansion of the model architecture. Owing to the design's foundation on the prevailing visual model architecture, our model is dynamically scalable and allows for the selection of different foundation model knowledge bases and visual backbones according to the application scenario.

Overall, we have utilized a framework based on a foundation model. It leverages its well-trained backbone to more effectively refine architectural feature representation, thereby achieving more precise task completion in the architecture-related domain. Additionally, our model possesses a strong capacity for knowledge, leading us to explore a unified paradigm for multitasking. Also, our model is pre-trained on large and diverse datasets, enabling it to learn rich feature representations. These pre-trained features can be transferred to tasks in specific domains, assisting in addressing issues with small datasets or lack of labeled data, and can perform well on other data even without further fine-tuning. Depending on the specific downstream application scenarios, researchers have the flexibility to choose from various transfer learning methods, including zero-shot direct application, few-sample linear probing, medium-sample parameter fine-tuning, etc.

However, our work still has some limitations. First, the size of our backbone is only at the ViT-L level, which means the precision advantage is not as pronounced as we would like it to be. In the future, training a backbone with a larger capacity is necessary, as it will provide better feature extraction and generalization capabilities. Second, our training data all come from open sources and have not undergone data cleaning, leading to substantial noise. The volume of data is also much less than that used to train foundation models in the natural image domain. Going forward, it will be necessary to collect and reconstruct data, which will significantly enhance the model's generality. Third, to accommodate the unification of heterogeneous tasks, specifically, the building extraction task involving single image input and the CD task involving double image inputs, we have duplicated the single building image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

While this method may seem inelegant and results in repeated calculations, it is currently a necessary compromise. Future research could investigate using the cross-attention mechanism to query varying quantities of image inputs dynamically. Finally, this article focused on buildings, addressing both building extraction and CD related to buildings. However, in the field of remote sensing image interpretation, there are many other objects of study with similar needs, such as roads, crops, and more. We hope that subsequent work can expand the data categories, enabling the model to perform better across the various key categories in remote sensing imagery.

V. Conclusion

In this article, we introduce a general building interpretation model, RSBuilding, developed from the perspective of the foundation model, which realizes the tasks of building extraction and building CD within a unified framework. Our objective is to move beyond meticulously designed branches or modules, and instead utilize a unified foundation model to enhance cross-scene generalization and task universality during building understanding. Specifically, we adhere to the mainstream transformer architecture by employing existing visual foundation models' encoders with general knowledge to extract image features. We have designed a straightforward multilevel feature sampler to enrich scale information. To unify task representation, we propose a cross-attention decoder based on task prompts. Regarding training data and strategies for the foundation model, we have addressed the current issue of limited datasets containing annotations for both tasks simultaneously, by developing a federated training strategy that enables the model to converge smoothly. This training strategy can enhance the complementarity between the two interpretation tasks. Experimental results demonstrate that RSBuilding can achieve state-of-the-art performance on multiple diverse source datasets and exhibits robust zero-shot generalization ability.

References

[1] K. Chen, Z. Zou, and Z. Shi, "Building extraction from remote sensing images with sparse token transformers," *Remote Sens.*, vol. 13, no. 21, p. 4441, Nov. 2021.

[2] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112, doi: 10.1109/TGRS.2024.3365825.

[3] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514, doi: 10.1109/TGRS.2021.3095166.

[4] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317, doi: 10.1109/TGRS.2024.3376673.

[5] K. Chen et al., "Time travelling pixels: Bitemporal features integration with foundation model for remote sensing image change detection," 2023, *arXiv:2312.16202*.

[6] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117, doi: 10.1109/TGRS.2024.3356074.

[7] S. Xu et al., "Learning 1-bit tiny object detector with discriminative feature refinement," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 1–11.

[8] S. Xu et al., "Q-DETR: An efficient low-bit quantized detection transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 3842–3851.

[9] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[10] A. Asokan and J. Anitha, "Change detection techniques for remote sensing applications: A survey," *Earth Sci. Informat.*, vol. 12, no. 2, pp. 143–160, Jun. 2019.

[11] L. Luo, P. Li, and X. Yan, "Deep learning-based building extraction from remote sensing images: A comprehensive review," *Energies*, vol. 14, no. 23, p. 7982, Nov. 2021.

[12] L. Wang, S. Fang, X. Meng, and R. Li, "Building extraction with vision transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625711.

[13] M. Ghanea, P. Moallem, and M. Momeni, "Building extraction from high-resolution satellite images in urban areas: Recent methods and strategies against significant challenges," *Int. J. Remote Sens.*, vol. 37, no. 21, pp. 5234–5248, Nov. 2016.

[14] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[15] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.

[16] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-Spatial Inf. Sci.*, vol. 26, no. 3, pp. 262–288, 2023.

[17] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[18] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[19] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.

[20] H. Chen et al., "Continuous cross-resolution remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623320, doi: 10.1109/TGRS.2023.3325829.

[21] K. Chen et al., "OvarNet: Towards open-vocabulary object attribute recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 23518–23527.

[22] K. Chen et al., "Continuous remote sensing image super-resolution based on context interaction in implicit function space," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702216, doi: 10.1109/TGRS.2023.3272473.

[23] K. Chen, W. Li, J. Chen, Z. Zou, and Z. Shi, "Resolution-agnostic remote sensing scene classification with implicit neural representations," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[24] B. Sirmacek and C. Unsalan, "Building detection from aerial images using invariant color features and shadow information," in *Proc. 23rd Int. Symp. Comput. Inf. Sci.*, Oct. 2008, pp. 1–5.

[25] Y. Zhang, "Optimisation of building detection in satellite images by combining multispectral classification and texture filtering," *ISPRS J. Photogramm. Remote Sens.*, vol. 54, no. 1, pp. 50–60, Feb. 1999.

[26] S.-H. Zhong, J.-J. Huang, and W.-X. Xie, "A new method of building detection from a single aerial photograph," in *Proc. 9th Int. Conf. Signal Process.*, Oct. 2008, pp. 1219–1222.

[27] Y. Li and H. Wu, "Adaptive building edge detection by combining LiDAR data and aerial images," *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. 37, no. B1, pp. 197–202, 2008.

[28] P. S. Tiwari and H. Pande, "Use of laser range and height texture cues for building identification," *J. Indian Soc. Remote Sens.*, vol. 36, no. 3, pp. 227–234, Sep. 2008.

[29] M. Awrangjeb, C. Zhang, and C. S. Fraser, "Improved building detection using texture information," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 38, pp. 143–148, Apr. 2013.

[30] C. Liu, R. Zhao, H. Chen, Z. Zou, and Z. Shi, "Remote sensing image change captioning with dual-branch transformers: A new method and a large scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5633520, doi: 10.1109/TGRS.2022.3218921.

[31] C. Yan et al., "VISA: Reasoning video object segmentation via large language models," 2024, *arXiv:2407.11325*.

[32] J. Wang and G. Kang, "Learn to rectify the bias of clip for unsupervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 4102–4112.

[33] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[34] S. Wei, S. Ji, and M. Lu, "Toward automatic building footprint delineation from aerial images using CNN and regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2178–2189, Mar. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

[35] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[36] Q. Zhu et al., "Knowledge-guided land pattern depiction for urban land use mapping: A case study of Chinese cities," *Remote Sens. Environ.*, vol. 272, Apr. 2022, Art. no. 112916.

[37] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," 2021, *arXiv:2105.15203*.

[38] Q. Zhu, Z. Li, T. Song, L. Yao, Q. Guan, and L. Zhang, "Unrestricted region and scale: Deep self-supervised building mapping framework across different cities from five continents," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 344–367, Mar. 2024.

[39] D. Wen et al., "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 68–101, Dec. 2021.

[40] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[41] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[42] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111, doi: 10.1109/TGRS.2023.3277496.

[43] C. Jia et al., "Scaling up visual and vision-language representation learning with noisy text supervision," in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 4904–4916.

[44] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[45] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," OpenAI, Tech. Rep., 2018, pp. 1–12.

[46] T. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 1877–1901.

[47] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.

[48] N. Ding et al., "Parameter-efficient fine-tuning of large-scale pre-trained language models," *Nature Mach. Intell.*, vol. 5, no. 3, pp. 220–235, Mar. 2023.

[49] L. Salewski, S. Alaniz, I. Rio-Torto, E. Schulz, and Z. Akata, "In-context impersonation reveals large language models' strengths and biases," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–14.

[50] L. Ouyang et al., "Training language models to follow instructions with human feedback," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 27730–27744.

[51] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[52] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 23716–23736.

[53] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.

[54] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[55] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[56] C. Liu, K. Chen, B. Chen, H. Zhang, Z. Zou, and Z. Shi, "RSCaMa: Remote sensing image change captioning with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[57] A. A. Aleissaee et al., "Transformers in remote sensing: A survey," *Remote Sens.*, vol. 15, no. 7, p. 1860, Mar. 2023.

[58] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[59] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.

[60] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[61] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[62] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[63] C. Pang, J. Wu, J. Ding, C. Song, and G.-S. Xia, "Detecting building changes with off-nadir aerial images," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140306.

[64] *XView2*. Accessed: Nov. 10, 2023. [Online]. Available: https://www.xview2.org/

[65] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[66] V. Mnih, "Machine learning for aerial image labeling," Ph.D. thesis, Univ. Toronto, Toronto, ON, Canada, 2013.

[67] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The Inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.

[68] AIcrowd. (2023). *Mapping Challenge*. Accessed: Nov. 10, 2023. [Online]. Available: https://www.aicrowd.com/challenges/mapping-challenge

[69] (2018). *2018 Open AI Tanzania Building Footprint Segmentation Challenge*. Accessed: Nov. 10, 2023. [Online]. Available: https://competitions.codalab.org/competitions/20100

[70] S. Holail, T. Saleh, X. Xiao, and D. Li, "AFDE-Net: Building change detection using attention-based feature differential enhancement for satellite imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[71] H. Li, F. Zhu, X. Zheng, M. Liu, and G. Chen, "MSCDUNet: A deep learning framework for built-up area change detection integrating multispectral, SAR, and VHR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5163–5176, 2022.

[72] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, Dec. 2021.

[73] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[74] M. Contributors. (2020). *MMSegmentation: OpenMMLab Semantic Segmentation Toolbox and Benchmark*. [Online]. Available: https://github.com/open-mmlab/mmsegmentation

[75] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[76] S. Seong and J. Choi, "Semantic segmentation of urban buildings using a high-resolution network (HRNet) with channel and spatial attention gates," *Remote Sens.*, vol. 13, no. 16, p. 3087, Aug. 2021.

[77] H. Zhang, Y. Liao, H. Yang, G. Yang, and L. Zhang, "A local–global dual-stream network for building extraction from very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1269–1283, Mar. 2022.

[78] Y. Liu, Z. Zhao, S. Zhang, and L. Huang, "Multiregion scale-aware network for building extraction from high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626310, doi: 10.1109/TGRS.2022.3192614.

[79] Y. Zhou et al., "BOMSC-Net: Boundary optimization and multi-scale context awareness based building extraction from high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618617.

[80] Z. Liu, Q. Shi, and J. Ou, "LCS: A collaborative optimization framework of vector extraction and semantic segmentation for building extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5632615, doi: 10.1109/TGRS.2022.3215852.

[81] L. Xu, Y. Li, J. Xu, Y. Zhang, and L. Guo, "BCTNet: Bi-branch cross-fusion transformer for building footprint extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402014, doi: 10.1109/TGRS.2023.3262967.

[82] H. Guo, X. Su, C. Wu, B. Du, and L. Zhang, "Decoupling semantic and edge representations for building footprint extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613116, doi: 10.1109/TGRS.2023.3287298.

[83] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

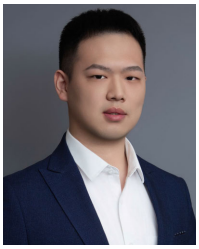
[84] A. Codegoni, G. Lombardi, and A. Ferrari, "TINYCD: A (Not So) deep learning model for change detection," 2022, *arXiv:2207.13159*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.

[85] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.

[86] P. Chen, B. Zhang, D. Hong, Z. Chen, X. Yang, and B. Li, "FCCDN: Feature constraint network for VHR image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 101–119, May 2022.

[87] K. Yang et al., "Semantic change detection with asymmetric Siamese networks," 2020, *arXiv:2010.05687*.



Mingze Wang received the B.E. degree from Beihang University, Beijing, China, in 2022, where he is currently pursuing the M.S. degree.

His research interests include deep learning, computer vision, and multimodal.



Lili Su received the B.E. degree from Northeast Forestry University, Harbin, China, in 2022. She is currently pursuing the master's degree in computer technology with Beijing University of Posts and Telecommunications, Beijing, China.

Her research interests include deep learning, reinforcement learning, edge computing, the Internet of Things (IoT), and communication network management.



Cilin Yan received the B.E. degree from Jilin University, Changchun, China, in 2022. He is currently pursuing the M.S. degree with Beihang University, Beijing, China. His research interests include deep learning, computer vision, and multimodal.



Sheng Xu received the B.E. degree in automotive engineering and the Ph.D. degree in pattern recognition and intelligent system from Beihang University, Beijing, China, in 2018 and 2024, respectively.

He is working as an Algorithm Expert at Alibaba Group, Beijing. He has made significant contributions to the field and has published about a dozen papers as the first author in top-tier conferences and journals such as Conference on Computer Vision and Pattern Recognition (CVPR), European Conference on Computer Vision (ECCV), International Conference on Computer Vision (ICCV), International Council for Machinery Lubrication (ICML), Conference on Neural Information Processing Systems (NeurIPS), Association for the Advancement of Artificial Intelligence, The British Machine Vision Conference (BMVC), International Journal of Computer Vision, and ACM Transactions on Multimedia Computing, Communications, and Applications (ACM TOMM), four of which are selected as oral or highlight presentations. He actively participates in the academic community as a Reviewer for various international journals and conferences, including CVPR, ECCV, ICCV, ICML, NeurIPS, International Conference on Learning Representations, and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT)/IEEE TRANSACTIONS ON INTELLIGENT VEHICLES (TIV). His expertise has also led to his group's victory in the ECCV 2020 tiny object detection challenge. His research interests include deep learning, computer vision, and pattern recognition.



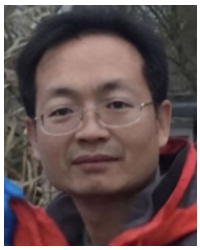
Pengcheng Yuan received the B.S. degree from the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2012, and the M.S. degree from the School of Computer Science and Engineering, Beihang University, Beijing, China, in 2019.

He is currently working as a Senior Engineer at Xiaohongshu Inc., Beijing.



Xiaolong Jiang received the B.S. degree from the School of Electrical and Information Engineering, Beihang University, Beijing, China, in 2012, the M.S. degree from the Fu Foundation School of Engineering and Applied Science, Columbia University, New York, NY, USA, in 2014, and the Ph.D. degree from the School of Electrical and Information Engineering, Beihang University, in 2019.

He is currently working as a Senior Engineer at Xiaohongshu Inc., Beijing.



Baochang Zhang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in computer science from Harbin Institute of the Technology, Harbin, China, in 1999, 2001, and 2007, respectively.

He is a Full Professor with the Institute of Artificial Intelligence, Beihang University, Beijing, China. He was selected by the Program for New Century Excellent Talents at the Ministry of Education of China, also selected as Academic Advisor of Deep Learning Lab of Baidu Inc., Beijing, and a Distinguished Researcher of Beihang Hangzhou Institute, Hangzhou, Zhejiang, China. He was a Research Fellow with the Chinese University of Hong Kong, PolyU, Hong Kong, and with Griffith University, Brisban, QLD, Australia. He was also a Post-Doctoral with the Italian Institute of Technology, Genoa, Italy. His HGPP and LDP methods were the state-of-the-art feature descriptors with 1169 and 734 Google Scholar citations. His group also received the ECCV 2020 tiny object detection, the COCO object detection, and ICPR 2020 Pollen recognition challenges. His research interests include deep learning, computer vision, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:50:30 UTC from IEEE Xplore. Restrictions apply.